```verilog
// ============================================================================
//   Ver  :| Author              :| Mod. Date :| Changes Made:
//   V1.1 :| Alexandra Du        :| 06/01/2016:| Added Verilog file
// ============================================================================


//=======================================================
//  This code is generated by Terasic System Builder
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`define ENABLE_ADC_CLOCK
`define ENABLE_CLOCK1
`define ENABLE_CLOCK2
`define ENABLE_SDRAM
`define ENABLE_HEX0
`define ENABLE_HEX1
`define ENABLE_HEX2
`define ENABLE_HEX3
`define ENABLE_HEX4
`define ENABLE_HEX5
`define ENABLE_KEY
`define ENABLE_LED
`define ENABLE_SW
`define ENABLE_VGA
`define ENABLE_ACCELEROMETER
`define ENABLE_ARDUINO
`define ENABLE_GPIO

module DE10_LITE_Golden_Top(

   //////////// ADC CLOCK: 3.3-V LVTTL //////////
`ifdef ENABLE_ADC_CLOCK
   input                      ADC_CLK_10,
`endif
   //////////// CLOCK 1: 3.3-V LVTTL //////////
`ifdef ENABLE_CLOCK1
   input                      MAX10_CLK1_50,
`endif
   //////////// CLOCK 2: 3.3-V LVTTL //////////
`ifdef ENABLE_CLOCK2
   input                      MAX10_CLK2_50,
`endif

   //////////// SDRAM: 3.3-V LVTTL //////////
`ifdef ENABLE_SDRAM
   output          [12:0]     DRAM_ADDR,
   output           [1:0]     DRAM_BA,
   output                     DRAM_CAS_N,
   output                     DRAM_CKE,
   output                     DRAM_CLK,
   output                     DRAM_CS_N,
   inout           [15:0]     DRAM_DQ,
   output                     DRAM_LDQM,
   output                     DRAM_RAS_N,
   output                     DRAM_UDQM,
   output                     DRAM_WE_N,
`endif
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   //////////// SEG7: 3.3-V LVTTL //////////
`ifdef ENABLE_HEX0
   output          [7:0]     HEX0,
`endif
`ifdef ENABLE_HEX1
   output          [7:0]     HEX1,
`endif
`ifdef ENABLE_HEX2
   output          [7:0]     HEX2,
`endif
`ifdef ENABLE_HEX3
   output          [7:0]     HEX3,
`endif
`ifdef ENABLE_HEX4
   output          [7:0]     HEX4,
`endif
`ifdef ENABLE_HEX5
   output          [7:0]     HEX5,
`endif

   //////////// KEY: 3.3 V SCHMITT TRIGGER //////////
`ifdef ENABLE_KEY
   input           [1:0]     KEY,
`endif

   //////////// LED: 3.3-V LVTTL //////////
`ifdef ENABLE_LED
   output          [9:0]     LEDR,
`endif

   //////////// SW: 3.3-V LVTTL //////////
`ifdef ENABLE_SW
   input           [9:0]     SW,
`endif

   //////////// VGA: 3.3-V LVTTL //////////
`ifdef ENABLE_VGA
   output          [3:0]     VGA_B,
   output          [3:0]     VGA_G,
   output                    VGA_HS,
   output          [3:0]     VGA_R,
   output                    VGA_VS,
`endif

   //////////// Accelerometer: 3.3-V LVTTL //////////
`ifdef ENABLE_ACCELEROMETER
   output                    GSENSOR_CS_N,
   input           [2:1]     GSENSOR_INT,
   output                    GSENSOR_SCLK,
   inout                     GSENSOR_SDI,
   inout                     GSENSOR_SDO,
`endif

   //////////// Arduino: 3.3-V LVTTL //////////
`ifdef ENABLE_ARDUINO
   inout          [15:0]     ARDUINO_IO,
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   inout                      ARDUINO_RESET_N,
`endif

   //////////// GPIO, GPIO connect to GPIO Default: 3.3-V LVTTL //////////
`ifdef ENABLE_GPIO
   inout          [35:0]     GPIO
`endif
);


wire [2:0]          column;
wire                row;

seq_counter #(
    .CLK_FREQ       (50000000),
    .UPDATE_HZ      (2)
) sequence_counter (
    .clk_i          (MAX10_CLK1_50),
    .rst_ni         (KEY[0]),
    .column_o       (column),
    .row_o          (row)
);

display display_output (
    .clk_i      (MAX10_CLK1_50),
    .rst_ni     (KEY[0]),
    .column_i   (column),
    .row_i      (row),
    .HEX0       (HEX0),
    .HEX1       (HEX1),
    .HEX2       (HEX2),
    .HEX3       (HEX3),
    .HEX4       (HEX4),
    .HEX5       (HEX5)
);
endmodule

```